Stochastic Processes

Homework 1: Foundations of Probability Theory

Lecture 1 | 20 problems | 100 points

Assigned reading

Sheldon M. Ross, *Introduction to Probability Models*, 10th edition.

- **Primary:** Chapter 1, Sections 1.1–1.6 (probability foundations, conditional probability, and independence).
- **Supporting:** Chapter 2, Sections 2.1–2.5 and 2.8 (random variables, distributions, moments, and limit theorems).

Instructions

Answer all 20 questions. Show your reasoning, state assumptions, and identify the probability rules or distributions you use. Give exact answers when practical; round numerical probabilities to four decimal places. You may use a calculator or Python to check arithmetic, but provide mathematical derivations. No solutions are included in this handout.

I. Probability Foundations — 12 points

Problem 1. Sample spaces and events [3 points]

A system has two components, each of which either works (W) or fails (F) during a test. Treat the components as distinguishable.

(a) Write the sample space.

(b) Define the events $A =$ "exactly one component fails" and $B =$ "the first component works" as sets.

(c) Find $A \cup B$, $A \cap B$, and A^c.

Problem 2. Axioms and complements [3 points]

Suppose $P(A) = 0.62$, $P(B) = 0.47$, and $P(A \cap B) = 0.25$.

(a) Find $P(A \cup B)$ and $P(A^c \cap B^c)$.

(b) Find $P(A \cap B^c)$.

(c) Explain why A and B cannot be mutually exclusive.

Problem 3. Counting outcomes [3 points]

Three fair six-sided dice are rolled independently.

(a) How many ordered outcomes are possible?

(b) Find the probability that no die shows a 6.

(c) Find the probability that at least one die shows a 6.

Problem 4. Ordered versus unordered sampling [3 points]

From a set of eight distinct sensors, three are chosen for inspection.

(a) Count the selections if inspection order matters.

(b) Count the selections if order does not matter.

(c) If exactly two sensors are defective, find the probability that a uniformly chosen unordered group of three contains both defective sensors.

II. Conditional Probability and Independence 16 points

Problem 5. Sequential sampling [4 points]

An urn contains 6 red, 4 blue, and 2 green balls. Two balls are drawn without replacement.

(a) Find the probability that both are red.

(b) Find the probability that the second is blue given that the first is red.

(c) Find the probability of one red and one blue in either order.

Problem 6. Total probability [4 points]

A factory uses lines A, B, and C to produce 50%, 30%, and 20% of its devices. Their defect rates are 1%, 3%, and 5%, respectively.

(a) Express the event "device is defective" using a partition by production line.

(b) Calculate the probability that a randomly selected device is defective.

(c) Explain why averaging the three defect rates without weights would be incorrect.

Problem 7. Bayesian inspection [4 points]

Continue Problem 6. An inspection flags a defective device with probability 0.96 and flags a nondefective device with probability 0.04.

(a) Find the probability that a randomly selected device is flagged.

(b) Given a flag, calculate the probability that the device is actually defective.

(c) Interpret the result in the context of the base rate.

Problem 8. Independence versus exclusion [4 points]

Let $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$.

(a) Determine whether A and B are independent; justify your answer.

(b) Determine whether they are mutually exclusive.

(c) Prove that two mutually exclusive events with strictly positive probabilities cannot be independent.

III. Random Variables and Distributions 20 points

Problem 9. A discrete probability law [5 points]

A random variable X has PMF $p_X(k) = c(k+1)$ for $k \in \{0, 1, 2, 3\}$ and zero otherwise.

(a) Determine c.

(b) Calculate $P(X \geq 2)$ and $P(X < 2)$.

(c) Write the CDF $F_X(x)$ for all real x.

Problem 10. Binomial reliability [5 points]

A system has five independent components, each operational with probability 0.9. Let X be the number operational.

(a) Identify the distribution of X and its parameters.

(b) Find $P(X \geq 4)$.

(c) Find $E[X]$ and $\mathrm{Var}(X)$.

Problem 11. Poisson arrivals [5 points]

Customers arrive at a help desk according to a Poisson process with rate 8 per hour. Let N be the number of arrivals in 30 minutes.

(a) Identify the distribution of N and its parameter.

(b) Find $P(N = 3)$ and $P(N \geq 1)$.

(c) Find the mean and standard deviation of N.

Problem 12. Exponential waiting times [5 points]

In the setting of Problem 11, let T be the time in hours until the next arrival.

(a) State the density and survival function of T.

(b) Find $P(T > 15 \text{ minutes})$.

(c) Find $P(5 < T \leq 15 \text{ minutes})$.

(d) Find $E[T]$ in minutes.

IV. Expectation, Covariance, and Joint Distributions 24 points

Problem 13. Expectation and variance from first principles [6 points]

Let X take values $-1, 0, 2$ with respective probabilities $1/4, 1/2, 1/4$.

(a) Compute $E[X]$ and $E[X^2]$ directly from the PMF.

(b) Compute $\mathrm{Var}(X)$ using $E[X^2] - E[X]^2$.

(c) Let $Y = 3X - 2$. Find $E[Y]$ and $\mathrm{Var}(Y)$ without rebuilding its PMF.

Problem 14. Dependent counts [6 points]

Two hourly arrival counts X and Y each have mean 10 and variance 10, but $\mathrm{Cov}(X, Y) = 3$.

(a) Find $E[X + Y]$ and $\mathrm{Var}(X + Y)$.

(b) Find $\mathrm{Var}(X - Y)$.

(c) Compare your variance in (a) with the result if X and Y were independent. Explain the difference.

Problem 15. Joint distribution [6 points]

The joint PMF of binary random variables $X, Y \in \{0, 1\}$ is given by

	$Y = 0$	$Y = 1$
$X = 0$	0.30	0.20
$X = 1$	0.10	0.40

(a) Find both marginal PMFs.

(b) Compute $P(X = 1 \mid Y = 1)$.

(c) Find $E[XY]$ and $\mathrm{Cov}(X, Y)$.

(d) Are X and Y independent? Justify.

Problem 16. Continuous densities [6 points]

A waiting-time variable has density $f_T(t) = ct(2 - t)$ for $0 \le t \le 2$, and zero otherwise.

(a) Find the normalizing constant c.

(b) Derive $F_T(t)$ for all real t.

(c) Compute $P(T > 1)$ and $E[T]$.

(d) Explain why $P(T = 1) = 0$ even though $f_T(1) > 0$.

V. Synthesis and Mathematical Reasoning 28 points

Problem 17. Pairwise versus mutual independence [7 points]

Two independent fair coins are tossed. Let $A =$ "first coin is heads," $B =$ "second coin is heads," and $C =$ "both coins show the same face."

(a) List the outcomes in each event.

(b) Show that every pair among A, B, C is independent.

(c) Determine $P(A \cap B \cap C)$ and compare it with $P(A)P(B)P(C)$.

(d) Explain what this demonstrates about pairwise and mutual independence.

Problem 18. A two-stage reliability model [7 points]

A device is selected from one of two suppliers: supplier 1 with probability 0.7, supplier 2 with probability 0.3. Its lifetime T (in hours), conditional on supplier i, is exponential with rate $\lambda_1 = 1/100$ or $\lambda_2 = 1/50$, respectively.

(a) Use total probability to derive $P(T > t)$ for $t \ge 0$.

(b) Calculate $P(T > 80)$.

(c) Given survival beyond 80 hours, find the probability the device came from supplier 1.

(d) Use conditional expectation to compute $E[T]$.

Problem 19. Law of large numbers and the CLT [7 points]

Let $X_1, \ldots, X_{400}$ be independent, identically distributed service times with $E[X_i] = 12$ minutes and $\mathrm{Var}(X_i) = 16$ minutes^2. Define $\overline{X} = \frac{1}{400} \sum_{i=1}^{400} X_i$.

(a) Find $E[\overline{X}]$ and $\mathrm{Var}(\overline{X})$.

(b) Use the central limit theorem to approximate $P(11.6 \leq \overline{X} \leq 12.4)$, expressing the result using the standard normal CDF Φ.

(c) Explain how the law of large numbers and central limit theorem answer different questions about $\overline{X}$.

Problem 20. Integrated service-center model [7 points]

A service center receives arrivals according to a Poisson process at rate 6 per hour. Each arriving customer independently requires an additional diagnostic check with probability 0.2. During a two-hour interval, let N be the total number of arrivals and D the number requiring a diagnostic check.

(a) Identify the distribution of N and find $E[N]$ and $\mathrm{Var}(N)$.

(b) Conditional on $N = n$, identify the distribution of D and calculate $E[D \mid N = n]$.

(c) Use the law of total expectation to find $E[D]$.

(d) Using the law of total variance $\mathrm{Var}(D) = E[\mathrm{Var}(D \mid N)] + \mathrm{Var}(E[D \mid N])$, find $\mathrm{Var}(D)$.

(e) Calculate $P(D = 0)$ by conditioning on N and summing the resulting series. Interpret your answer.

Before you submit. Check that you have labeled all 20 problems, justified your calculations, kept time units consistent, and interpreted probabilities where requested.